Inequality of Opportunity in Territorial Financial Access: Evidence from Peru, 2002–2025

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Abstract

Peru's banking agents reached 95.5% of districts by 2025, yet only 28.5% have a full-service financial office. I apply the Human Opportunity Index (HOI) framework to 1,873 Peruvian districts observed annually from 2002 to 2025, estimating the HOI and its underlying D-index and decomposing the HOI penalty via Shapley values across six territorial circumstances: poverty, altitude, density, remoteness, natural region, and indigenous-language share. Under the office definition (access requires a financial office), the HOI moves slowly from 0.131 in 2002 to 0.202 in 2025, with all the gain from coverage expansion (19.5% to 28.5%); the D-index behind it stays essentially flat at ≈ 0.30 (0.326, 0.304, 0.291), so inequality of opportunity in office access did not budge in 24 years. Under the any-access definition (office or banking agent), the HOI jumps from 0.131 in 2002 to 0.937 in 2025, driven jointly by near-universal coverage (19.5% to 95.5%) and a collapse in the D-index (0.326 in 2011 to 0.019 in 2025) as agents fill in the territory. Density is the largest contributor to the office HOI penalty (28.8%), followed by poverty (20.5%), natural region (16.4%), altitude (14.3%), remoteness (11.4%), and indigenous language (8.7%); a geography-only specification reproduces 77.8% of the penalty. Agents equalized basic transactional access, but full-service infrastructure remained concentrated in denser, less-poor, lower-altitude territories throughout the two-decade period.

Keywords: inequality of opportunity; financial inclusion; banking deserts; Human Opportunity Index; D-index; Peru.
JEL codes: D63, G21, O16, R12.

1 Introduction

Formal finance shapes economic mobility, but only where households and firms can physically reach it. In countries with large geographic, ethnic, and income disparities, the place where people live can determine whether basic financial services are available at all [Galor and Zeira, 1993, Beck et al., 2007, Demirguc-Kunt and Levine, 2009]. Peru, with coastal cities, Quechua-speaking Andean highlands above 4,000 meters, and Amazonian settlements reachable only by river [Iguíñiz

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Echeverría, 1998], offers a sharp case for asking how unequally territorial financial infrastructure is distributed, and whether the recent expansion of banking agents has closed that gap.

Between 2002 and 2025, Peru's financial-office footprint[1] expanded from 365 to 533 districts covered by at least one office of any regulated institution (19.5% to 28.5% of Peru's 1,873 districts). Over the same window, any-access coverage, defined as at least one office or banking agent, rose from 365 to 1,788 districts (19.5% to 95.5%), a near-universal expansion driven almost entirely by agents. Agents spread rapidly into bodegas, pharmacies, and retail outlets under the SBS agent-banking framework. Yet a full-service office and a shopkeeper-run agent are not equivalent: agents process deposits, withdrawals, and bill payments, but not credit evaluation, term deposits, or advisory services. Peru's decade of financial-inclusion progress is therefore not a single story but two: a rapid closing of the extensive-margin gap on the agent side, and an office-side frontier that stopped moving after 2011.

I use this two-tier setting to measure inequality of opportunity (IOp) in territorial financial access. Following Roemer [1998] and Roemer and Trannoy [2016], IOp analyses distinguish inequality attributable to circumstances beyond individual control from inequality attributable to effort. Because the outcome studied (whether a district hosts a regulated financial access point) does not admit a well-defined effort dimension at the district level, I adopt the ex-ante approach [Fleurbaey and Peragine, 2013], in which inequality of opportunity is measured as the variation in expected access across circumstance-defined groups without requiring observation of effort [Ferreira and Gignoux, 2011]. The IOp measure I use is the D-index, a dissimilarity measure for dichotomous outcomes adapted to the IOp setting by Paes de Barros et al. [2009], who apply it to groups defined by predetermined circumstances. I also report the composite Human Opportunity Index (HOI) proposed in the same paper, which discounts average coverage by the D-index; following Brunori et al. [2013], I read the HOI as a development index sensitive to IOp rather than as a measure of IOp per se. The framework was originally developed for basic services such as water, sanitation, electricity, and schooling [Paes de Barros et al., 2009], extended to a broader LAC setup by Molinas Vega et al. [2010], and applied cross-country by Brunori et al. [2013]. The D-index is computed in two steps. First, a logit maps a set of predetermined circumstances into a predicted access probability for each unit. Second, the D-index summarizes how unequally those predicted probabilities are distributed across circumstance-defined groups: it takes the value 0 when access is independent of circumstances and rises toward 1 as it becomes concentrated in advantaged groups. The composite HOI combines the resulting D-index with average coverage as $\text{HOI} = \bar{p}\,(1 - D)$. A Shapley decomposition then attributes the HOI penalty $P = \bar{p} \cdot D$ (equivalently the D-index, since the two shares coincide up to a $\bar{p}$ rescaling) to individual circumstances, so the total is split into interpretable shares.

[1] Throughout, "financial office" refers to any office of a regulated financial institution: commercial banks, Banco de la Nación (state bank), CMACs (municipal savings banks, *Cajas Municipales de Ahorro y Crédito*), CRACs (rural savings banks, *Cajas Rurales de Ahorro y Crédito*), EDPYMEs (small-enterprise development entities, *Entidades de Desarrollo para la Pequeña y Microempresa*), and financieras. "Banking agent" (*cajero corresponsal*) is a POS-based access point hosted in a retail establishment of any regulated institution.

Two considerations justify treating territorial financial access as an opportunity. First, Peruvian policy already treats it as one: the National Financial Inclusion Strategy [Gobierno del Perú, 2015] and the National Financial Inclusion Policy [Gobierno del Perú, 2019] make territorial access an explicit objective alongside water, education, and health, and Peru has ranked consistently among the leading Latin American environments in the EIU's Global Microscope [Economist Intelligence Unit, 2020], so any persistent territorial gap cannot easily be attributed to weak regulation. Second, a body of causal evidence [Burgess and Pande, 2005, Bruhn and Love, 2014, Nguyen, 2019] shows that physical bank presence has measurable effects on rural poverty and small-business activity, which supports reading access as a welfare-relevant precondition rather than a purely commercial variable.

Formal financial infrastructure fits this framework (binary outcome, predetermined circumstances, observable groups), and to my knowledge has not been analyzed with the HOI framework before.

The absence of an effort dimension in HOI is not a limitation here, in the same sense that Paes de Barros et al. [2009] argue it is not a limitation when the outcome concerns children. I treat six district-level features as circumstances: poverty, altitude, density, remoteness, natural region, and indigenous-language share.

I compute the HOI and its underlying D-index under two access definitions: an "office" definition (a district has access if it hosts at least one financial office) and an "any-access" definition (access is granted by either an office or a banking agent). Under the office definition the HOI moves slowly from 0.131 in 2002 to 0.202 in 2025, all from coverage expansion (19.5% to 28.5%): the D-index stays essentially flat at ≈ 0.30 (0.326 in 2002, 0.291 in 2025), so circumstances predict office presence today about as sharply as they did over two decades ago. Under the any-access definition the HOI jumps from 0.131 in 2002 to 0.937 in 2025 as agents fill in the territory: coverage rises from 19.5% to 95.5% and the D-index collapses from 0.326 in 2011 to 0.019 in 2025, so circumstances no longer predict whether a district has any point of financial contact. The two outcomes together show that agents equalized access to basic transactions, but did not change who has access to full-service infrastructure. A Shapley decomposition of the 2025 office HOI penalty attributes the largest shares to density (28.8 percent) and poverty (20.5 percent), consistent with a market-driven office frontier. As a robustness exercise, restricting the logit to only the three geographic circumstances (altitude, remoteness, natural region) yields an HOI penalty equal to 77.8 percent of the full-model penalty: geography alone accounts for roughly four-fifths of the measured inequality. Two counterfactual exercises, run for every year of the panel, frame the policy discussion (Figure 3). The first removes state entities (Banco de la Nación and Agrobanco) from the office panel: without them, the 2025 office HOI would fall by 35 percent (from 0.202 to 0.132); the gap was much larger in the early 2000s (2002: 0.131 vs 0.030), reflecting the small size of the private banking system in that period, and has stabilized at around 0.07 HOI points since 2011. This quantifies the persistent redistributive role of BN. The second scenario closes the remaining desert districts by opening one office in each; the resulting HOI gain has fallen sharply as the agent network expanded, from 0.72 HOI points

in 2011 (with over a thousand remaining deserts) to 0.05 HOI points in 2025 (with fewer than a hundred). The window for aggressive desert-closing policy has largely closed.

This paper extends the HOI/D-index framework in two ways relative to the canonical setup [Paes de Barros et al., 2009, Molinas Vega et al., 2010]. First, the unit of analysis is the administrative district, an aggregated territorial unit, rather than the individual on which the canonical HOI is defined; the outcome (whether a district hosts a regulated financial office or agent) and the circumstances are therefore also defined at the district level. Second, the moral patient whose opportunity is being measured is the district resident, distinct from the unit of analysis; the framework's individual-level interpretation is preserved to the extent that geographic circumstances (natural region, altitude, remoteness) are perfectly shared within each district, and the geography-only decomposition in Section 5 quantifies how much of the exercise survives under that stricter reading.

The paper contributes to two literatures. To the IOp literature, it extends the HOI/D-index framework to financial infrastructure and to a new unit of analysis: the district as a carrier of territorial circumstances. It documents how coverage and inequality of opportunity evolve when a mature office network coexists with a fast-growing agent network. To the financial-inclusion and banking-deserts literature, it reframes the descriptive question from "how much access exists" to "how equally is it distributed across territories with different circumstances," and traces how that distribution shifts through a decade of agent expansion.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the empirical framework. Section 5 reports results, robustness checks, and policy counterfactuals. Section 6 discusses and concludes.

2 Related Literature

The paper builds on three literatures. It adopts the HOI/D-index framework as its measurement tool (drawing on the broader inequality-of-opportunity literature), extends it to a new unit of analysis (the administrative district) and a new class of outcomes (physical financial infrastructure), and situates the outcome as welfare-relevant using the empirical literature on physical financial access and banking deserts.

Inequality of opportunity. Applied IOp research has developed methods for decomposing outcome inequality into circumstance- and effort-driven components, most of it on continuous outcomes such as earnings. Applications have extended beyond labor earnings to education [Ferreira and Gignoux, 2014], subnational analysis in high-inequality LAC contexts [Núñez and Tartakowsky, 2011], and aggregate growth [Marrero and Rodríguez, 2013]; the usefulness of IOp as a policy construct remains debated [Kanbur and Wagstaff, 2016, Hufe et al., 2018]. Two features of this literature matter for the present paper. First, ex-ante approaches compare expected outcomes across circumstance-defined groups, while ex-post approaches compare individuals at similar effort levels [Fleurbaey and Peragine, 2013, Ramos and Van de gaer, 2016]; because effort is not defined at the district level, this

paper adopts the ex-ante approach. Second, Ferreira and Gignoux [2011] show that in continuous-outcome IOp applications, estimates based on a subset of observed circumstances constitute a lower bound on the true IOp. This property does not extend automatically to binary outcomes, so the D-index reported below should be read as a descriptive measure of circumstance-driven variation in access, informative about the six observed circumstances but not a formal lower bound on total IOp. Almost all this work is defined at the individual or household level; the territorial application here shifts the unit of observation to the district.

The Human Opportunity Index. Paes de Barros et al. [2009] developed the HOI for binary basic-service access in Latin America, combining average coverage with the D-index. The canonical application takes the individual child (ages 0–16) as the unit of analysis and household attributes as circumstances, with outcomes in water, sanitation, electricity, and schooling; Molinas Vega et al. [2010] extend the framework to more LAC countries under the same child-focused setup, and Brunori et al. [2013] compute the HOI for 39 developing countries, interpreting it as a development index sensitive to IOp rather than a pure IOp measure. To my knowledge, the HOI has not been applied to territorial financial-access infrastructure. This paper is the first such application, using a Peruvian district-year panel (2002–2025) that covers all regulated financial institutions.

Physical financial access and banking deserts. A causal literature establishes that physical bank presence affects poverty, income, and small-business activity. Burgess and Pande [2005] exploit India's 1977–1990 branch-licensing rule and show that opening branches in rural unbanked locations significantly reduced rural poverty. Bruhn and Love [2014] exploit the simultaneous opening of over 800 Banco Azteca branches in Mexico and find a 7 percent increase in income and higher informal-business activity. Suri and Jack [2016] document a comparable mechanism through mobile-money agents in Kenya: proximity to M-PESA agents raised per-capita consumption and lifted an estimated 194,000 households (2 percent of Kenyan households) out of extreme poverty. Two US studies mirror these results at the margin: Nguyen [2019] shows that branch closings reduce local small-business lending by $453,000 per closing, dissipating within six miles, and Célerier and Matray [2019] exploit US interstate branching deregulation to show that exogenous branch expansion increases financial inclusion of low-income households. A parallel literature documents the uneven geography of physical bank access, mostly in the United States [Hegerty, 2016, Van Leuven et al., 2024], with cross-country evidence on branch and ATM penetration [Beck et al., 2007]. This literature operationalizes access through physical-distance thresholds rather than administrative boundaries, and documents spatial gaps without asking how much is attributable to predetermined circumstances, the question at the core of the IOp framework.

3 Data

The empirical dataset is a district-year panel for Peru built from administrative data reported by Peru's financial supervisory authority, the Superintendencia de Banca, Seguros y AFP [SBS;

Superintendencia de Banca, Seguros y AFP del Peru, 2026], which supervises all regulated financial institutions under Peru's General Law of the Financial System [Congreso de la República del Perú, 1996]. The SBS data cover *all* regulated financial institutions: commercial banks (*banca múltiple*), municipal savings banks (*cajas municipales de ahorro y crédito*, CMACs), rural savings banks (*cajas rurales*, CRACs), small-enterprise development institutions (EDPYMES), finance companies (*financieras*), and state entities Banco de la Nación and Agrobanco. The panel spans 2002–2025, using SBS annual reports for private and specialized institutions and the Banco de la Nación institutional directory for its own offices. The CMACs (such as Caja Huancayo, Caja Arequipa, and Caja Piura) and Banco de la Nación are particularly important for territorial reach, operating in many highland and rural districts where commercial banks are absent. Municipal ownership of the CMACs matters for the interpretation: each is historically anchored to its home municipality (Caja Huancayo to Huancayo, Caja Arequipa to Arequipa, and so on), which explains their strong secondary-city and provincial presence relative to what a private, Lima-headquartered lender would build.

Two binary outcomes are constructed. The first, **financial office**, equals one if district d has at least one office of any regulated financial institution in year t. The second, **any access**, equals one if the district has at least one financial office *or* banking agent (*cajero corresponsal*) in year t. A district with neither an office nor an agent is labeled a **financial desert**. Banking agents are POS-based access points hosted in retail establishments under an SBS regulatory framework; district-level agent counts come from the SBS annual inclusion-finance reports. They can process deposits, withdrawals, loan disbursements and repayments, utility payments, and government transfers (including the Juntos conditional cash transfer program and the Pensión 65 non-contributory pension), but do not typically provide account opening, credit evaluation, safe-deposit services, or complex advisory products. Agents are operated either directly by each regulated institution or through multi-institution aggregators that host multiple institutions' transaction rails at a single physical location.

Six district-level circumstance variables are merged from official sources. The *poverty rate* comes from the 2018 INEI district poverty map. *Altitude* (meters above sea level, of the district capital) comes from INEI district records. *Population density* is computed from 2017 census population and district area. *Remoteness* is the Haversine (great-circle) distance from each district to the most populous district of its department (2017 census population). *Natural region* follows INEI's standard classification, aggregated to Coast, Sierra, and Amazon. *Indigenous-language share* is the proportion of individuals aged 3 or older whose mother tongue is a Peruvian indigenous language (Quechua, Aymara, or any Amazonian native language), computed from Peru's XII Population Census 2017.

The office count captures offices with a distinct *código de oficina* in the source reports. Two sources of financial presence are not captured in the panel: (i) *cooperativas de ahorro y crédito* (COOPAC), brought under SBS supervision in 2018 but not yet integrated into the district-level office panel; and (ii) ATMs, whose district coverage is a subset of the agent network, and mobile-

banking access, which presupposes an existing account. Banco de la Nación "Oficinas Especiales" (points of service without deposit or credit balances) *are* captured, because the BN institutional directory used here includes them explicitly.

The main estimation sample contains 1,873 districts and 44,952 district-year observations over 2002–2025. Coverage matters: in 2025, only 257 districts have a commercial bank office, but 533 have at least one office when all regulated institutions are included; 549 districts hosted a regulated office at some point during the panel and 1,324 never did. Appendix Table A2 compares the two groups: the ever-served districts are systematically less poor, less indigenous, denser, at lower altitude, closer to the departmental population hub, and more likely to be in the Coast, with the largest gaps in density, altitude, and Sierra-region share. Because these circumstance variables are time-invariant, this static contrast summarizes the two-decade panel and foreshadows the plateau documented in Section 5.

4 Empirical Framework

The empirical strategy combines three elements: a logit model that maps district circumstances into predicted access probabilities, the D-index that summarizes how unequally those probabilities are distributed, and a Shapley decomposition that attributes each circumstance's contribution to the resulting HOI penalty.

The empirical strategy is guided by a simple two-tier logic of financial-infrastructure location. A full-service office requires premises, security, and staff, so it is only viable where scale, income, and geographic accessibility are jointly favorable. A banking agent is a POS terminal hosted in an existing retail establishment; its fixed cost is much lower, so it can operate in smaller and poorer districts. Financial deserts are districts where neither threshold is met, i.e., jointly small in population, low in income, and constrained by geography. The Shapley decomposition below quantifies how much each observed circumstance contributes to office-level territorial inequality.

Access model. For each year in the main window (2002 through 2025, annual), the probability that district d has the outcome of interest (a financial office, or any regulated access point) is modeled as a logit:

$$\Pr(Y_{dt} = 1 \mid C_d) = \Lambda(\alpha_t + C_d'\beta_t), \tag{1}$$

where Y_{dt} is the binary outcome, C_d is the vector of observed circumstances, and $\Lambda(\cdot)$ is the logistic cumulative distribution function. The circumstance vector includes poverty rate, log altitude, log density, log distance to the departmental hub, and indigenous-language share (all standardized to mean zero, unit variance across the pooled panel), plus natural-region dummies (Coast and Amazon, with Sierra as the omitted category). The estimated probability $\hat{p}_{dt}$ represents the predicted access probability associated with district d's circumstances in year t.

The logit is estimated separately for each year rather than pooled across time. This allows the coefficients, and thus the mapping from circumstances to predicted probabilities, to evolve.

The logit serves an instrumental role: its coefficients are not interpreted causally, and the model's purpose is to generate predicted probabilities that feed into the D-index.

D-index and Human Opportunity Index. The D-index is a dissimilarity measure for dichotomous outcomes. Paes de Barros et al. [2009] adapt it to the IOp setting by defining groups through predetermined circumstances, and combine it with average coverage into the Human Opportunity Index, $\text{HOI} = \bar{p}\,(1 - D)$. Following Paes de Barros et al. [2009], I implement the *parametric* version, in which the group-specific access probabilities are the logit-fitted values $\hat{p}_{dt}$ from equation (1) rather than empirical group means. Molinas Vega et al. [2010] apply the same specification and extend it to within-country subnational analysis and to comparisons with developed countries.

Let $\bar{p}_t$ denote average access coverage in year t and w_d district weights summing to one. The D-index is:

$$D_t = \frac{1}{2\bar{p}_t} \sum_d w_d \, |\hat{p}_{dt} - \bar{p}_t|. \tag{2}$$

Intuitively, D_t measures the fraction of access opportunities that would need to be reallocated across circumstance groups to achieve equal access. If all groups have the same predicted probability, $D = 0$. If access is perfectly sorted by circumstances, D approaches its maximum. The Human Opportunity Index combines coverage with this equity component:

$$\text{HOI}_t = \bar{p}_t(1 - D_t) = \bar{p}_t - P_t, \qquad P_t \equiv \bar{p}_t \cdot D_t. \tag{3}$$

Following Molinas Vega et al. [2010], P_t is the *penalty* for inequality of opportunity: the mass of covered opportunities that would need to be reallocated across circumstance groups to achieve equal access. The HOI rises when coverage increases *and* when the penalty falls. This distinction is central to the paper: aggregate coverage statistics cannot differentiate between broad-based expansion and expansion concentrated in already-advantaged territories. The D-index, and equivalently the penalty, can.

Two weighting schemes capture distinct normative objects. In the *unweighted* specification ($w_d = 1/N$), each territory is treated as an opportunity unit. This perspective asks: does the financial system reach all of Peru's administrative geography equally? A small district of 500 inhabitants and a large district of 500,000 receive equal weight. In the *population-weighted* specification ($w_d \propto$ district population, with the logit also estimated using population weights for consistency), each person is the unit. This asks: does the probability of living in a served district depend on circumstances? Because the paper's central question concerns territorial inequality of opportunity, I use the unweighted D-index and HOI as the baseline throughout the main results and report the population-weighted versions as a robustness check in Appendix Table A3 and Figure A1.

Shapley decomposition. To identify which circumstances contribute most to the penalty $P_t = \bar{p}_t \cdot D_t$, I use a Shapley-value decomposition, following the framework of Shorrocks [2013] and the applied implementation of Wendelspiess Chavez Juarez and Soloaga [2014]. Let S denote a subset

of circumstances and $P(S) = \bar{p} \cdot D(S)$ the penalty obtained from a logit model using only the circumstances in S (the logit's predicted mean equals the observed mean by construction, so $\bar{p}$ does not depend on S; only $D(S)$ varies). The Shapley contribution of circumstance k is:

$$\phi_k = \sum_{S \subseteq K \setminus \{k\}} \frac{|S|!(|K| - |S| - 1)!}{|K|!} \left[P(S \cup \{k\}) - P(S) \right], \tag{4}$$

where $|K|$ is the total number of circumstance groups. Each circumstance's contribution is its average marginal effect on the penalty across all possible subsets of other circumstances. The Shapley decomposition has two properties that matter for interpretation. First, it is *exact*: individual contributions sum to the full penalty (equivalently, $\sum_k \phi_k = \bar{p} \cdot D$). Second, it is *path-independent*: the attribution does not depend on the order in which variables are entered into the decomposition. Because $\bar{p}$ is constant across subsets, the relative shares of the penalty and of the D-index coincide; the penalty framing simply expresses each contribution in the same units as the HOI shortfall.

With six circumstance groups, the computation requires estimating $2^6 = 64$ logit models per year. Natural region enters as two dummies (Coast and Amazon) but is treated as a single circumstance group in the Shapley decomposition: its contribution is computed by adding or removing both dummies jointly. This grouping avoids inflating the degrees of freedom attributed to a single conceptual variable. Because the six circumstance variables are time-invariant (poverty from the 2018 map, density from the 2017 census, geography and language likewise fixed), year-to-year changes in Shapley shares reflect changes in the logit coefficients (how the mapping from circumstances to access probabilities evolves), not changes in the underlying distribution of circumstances.

When circumstances are correlated, individual Shapley contributions reflect both direct effects and shared variance. In the present application, density and remoteness are correlated at -0.60, and altitude and indigenous language at 0.45. The Shapley values are well-defined regardless of correlation (they provide a unique, fair attribution), but each circumstance's share partly reflects variance shared with correlated ones. To isolate the contribution of variables fully exogenous to market dynamics, I complement the six-circumstance specification with a geography-only robustness (altitude, remoteness, natural region), separate from potentially endogenous circumstances (poverty, density, indigenous language). Section 5 reports the result.

5 Results

The logit that feeds the HOI and D-index is estimated year by year over 2002–2025. Its fit is stable throughout the window (pseudo-R^2 around 0.18, AUC around 0.78). The dominant predictors of office presence are density (positive) and poverty (negative), consistent with a market-size logic of branch location. Amazon districts have a positive coefficient relative to Sierra, likely reflecting the hub role of Amazon provincial capitals. Coefficients and standard errors for every year of the panel are reported in Appendix Tables A4 and A5.

Figure 1 shows the annual evolution of the HOI (Panel A) and the underlying D-index (Panel

B) under the two access definitions across 2002–2025; full annual series including coverage are in Appendix Table A1. On the office side, the HOI moves slowly from 0.131 in 2002 to 0.202 in 2025. All 0.07 HOI points of the gain come from coverage expansion (19.5 percent in 2002 to 28.5 percent in 2025); the D-index behind it stays essentially flat at ≈ 0.30 (0.326 in 2002, 0.304 in 2011, 0.291 in 2025). This slow HOI progress is the central finding on the office side: the HOI advanced almost entirely along the extensive margin (adding coverage in already-favored circumstance groups), with essentially no equity improvement over 24 years, consistent with a mature banking sector operating close to the exhaustion of commercially viable office locations from very early in the period.

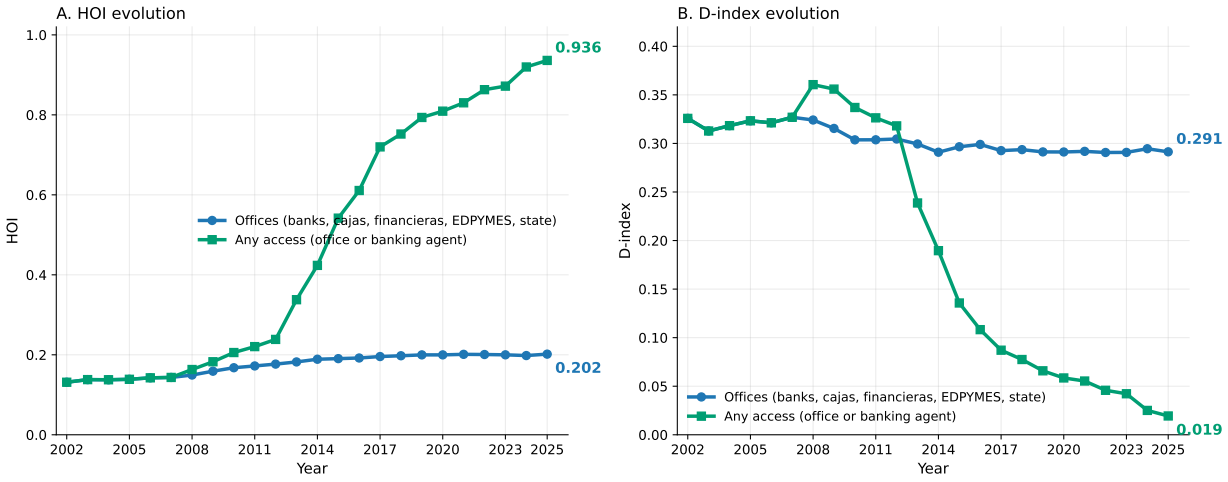


Figure 1: Annual evolution of the HOI (Panel A) and D-index (Panel B) under the two access definitions, 2002–2025. Full time series in Appendix Table A1.

Banking agents transform the picture on the any-access outcome. In 2011 the agent channel was still modest: any-access coverage stood at 32.7 percent (8 pp above office coverage) and the any-access HOI at 0.220 tracked the office HOI of 0.172 closely, with both D-index values in the 0.30–0.33 range. Over the following fourteen years the two diverge dramatically. By 2025 the any-access HOI hits **0.937**, driven jointly by near-universal coverage (**95.5 percent**) and a D-index collapse to **0.019**: basic access is distributed almost equitably across territories.

The composition of access shifted correspondingly. In 2002, 80.5 percent of districts were financial deserts and 19.5 percent had a financial office; banking agents did not yet exist as a regulated channel until 2008. By 2011, 24.7 percent of districts had an office, 8.0 percent had a banking agent but no office, and 67.3 percent remained deserts. By 2025 the landscape has reversed: 28.5 percent are served by an office, 67.0 percent have a banking agent but no office, and only 4.5 percent (85 districts) remain deserts. Geographically, offices concentrate along the coast and in departmental capitals, agents fill the intermediate Sierra and Amazon districts, and the remaining deserts cluster in the highest-altitude and most remote territories.

This two-tier pattern is the paper's central empirical finding. It means that Peru's financial infrastructure expansion largely resolved the *extensive margin* of basic access (the question of

whether a district has any financial service point at all), but left the *intensive margin* of service quality unresolved.

The Shapley decomposition is applied to the office HOI penalty, the outcome that carries the paper's central puzzle: the any-access D-index has collapsed to 0.019 by 2025, leaving little to decompose, while the office D-index persists at ≈ 0.29. The composition of the office decomposition is remarkably stable across 2002–2025 (Figure 2). Density is the largest contributor at around 29 percent, followed by poverty (around 21), natural region (16), altitude (14), remoteness (11), and indigenous language (9). Density and poverty jointly account for roughly half of the office penalty in every year, consistent with the commercial logic of office location: financial institutions (whether commercial banks, CMACs, or financieras) face fixed costs that require a minimum customer base to cover, so districts below this threshold do not attract offices regardless of other characteristics. Natural region reflects ecological and institutional differences across Peru's three macro-regions, with Sierra and Amazon districts facing higher operational costs than the coast. The indigenous-language coefficient in the logit is small and not individually significant, but the Shapley procedure captures its contribution through correlation with poverty, altitude, and remoteness.

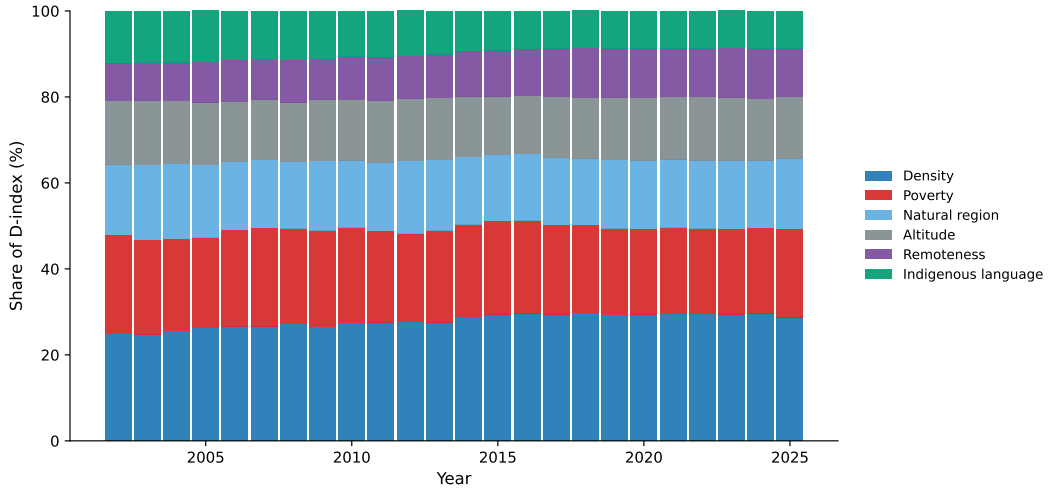


Figure 2: Shapley decomposition of the office HOI penalty over time, 2002–2025. Relative shares are identical to the D-index decomposition, since $P = \bar{p} \cdot D$.

The any-access decomposition (Appendix Figure A2) reveals a different pattern, though it must be read with caution: when the total D-index collapses to 0.019, absolute contributions become small and share-based comparisons are correspondingly noisy. The two largest qualitative changes between the office and any-access decompositions are that poverty's share collapses from roughly one-fifth of the office HOI penalty to a small fraction of the any-access penalty (unlike offices, agents do not select on poverty) and that indigenous language's share roughly doubles. Density's share remains dominant in relative terms in both, consistent with the fact that even lightweight agent infrastructure has a minimum-viable-scale requirement: bodegas and pharmacies host agents where enough traffic exists. Agents, with much lower fixed costs (a POS terminal in an

existing retail establishment), operate at scales offices cannot. The residual any-access inequality is concentrated in the few remaining districts that are simultaneously small, remote, high-altitude, and disproportionately indigenous-speaking: the intersection of scale, geography, and language, rather than any single circumstance.

The results are robust to alternative specifications. Under population weighting (Table A3 and Figure A1), the office HOI rises to 0.693 (D-index falls to 0.130) and the any-access HOI to 0.994 (D-index to 0.003), with population-weighted any-access coverage of 99.6 percent. The population-weighted version answers a different question than the territorial baseline: it asks how the probability that a randomly drawn Peruvian lives in a served district depends on territorial circumstances, and is conceptually closer (though still not identical) to the child-level HOI benchmarks reported for LAC basic services [Paes de Barros et al., 2009, Molinas Vega et al., 2010], since it aggregates over persons rather than territories. The territorial baseline (the unweighted HOI and D-index reported in Table A1) treats each district as an opportunity unit and is the natural object for the paper's central question about territorial inequality. A geography-only specification using only altitude, remoteness, and natural region (variables that are fully exogenous and unaffected by reverse causality) yields an office HOI penalty equal to 77.8 percent of the full-model penalty in 2025 (D-index 0.227 vs 0.291); the ratio remains between 75 and 80 percent in every year of the 2002–2025 window. Hard geography alone accounts for the bulk of the measured inequality of opportunity.

Two counterfactual simulations translate these findings into policy magnitudes (Figure 3, Appendix Table A6). The no-state-entity counterfactual removes BN and Agrobanco from the office panel: their absence would drop the 2025 office HOI from 0.202 to 0.132, reflecting both a lower coverage rate and a higher D-index (from 0.291 to 0.382). The HOI gap between observed and no-state was largest in the early 2000s (2002: 0.131 vs 0.030, when the private banking system was small) and has stabilized at around 0.07 HOI points since 2011. This quantifies BN's ongoing redistributive role: without state entities operating under the *Única Oferta Bancaria* mandate, territorial HOI would fall by 35 percent. The desert-elimination counterfactual forces has_office = 1 in every district without any access point. It shows sharply diminishing returns as the agent network expanded: in 2011 (with 1,260 remaining deserts), closing all deserts would have pushed the office HOI from 0.172 to 0.894, a gain of 0.72 HOI points; by 2025 (with 85 residual deserts), the same policy adds only 0.05 HOI points (from 0.202 to 0.256). The policy margin for aggressive desert closure has been largely absorbed by the extensive-margin expansion of agents.

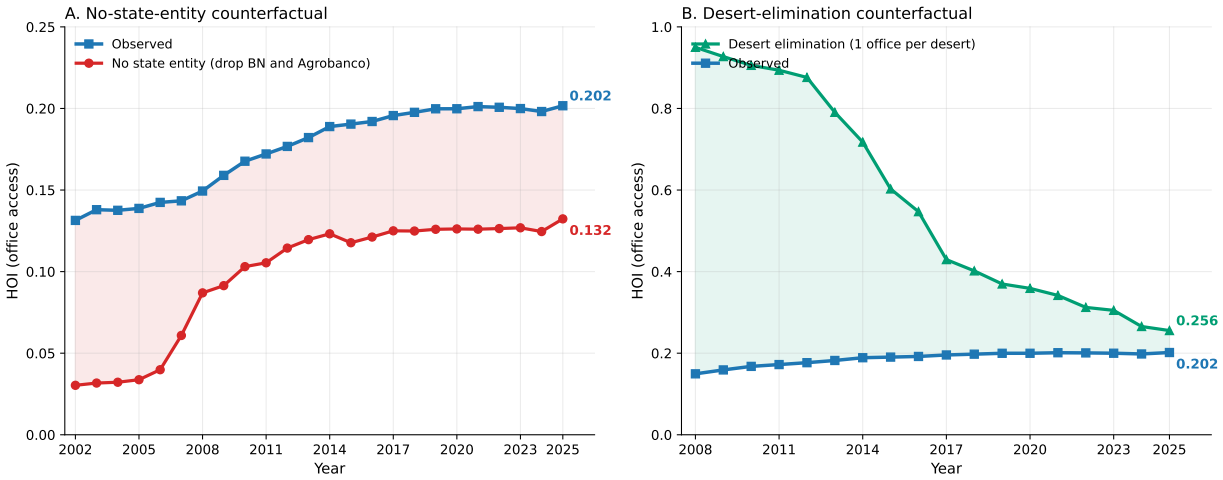


Figure 3: Policy counterfactuals on the office HOI. Panel A: no-state-entity (BN and Agrobanco removed), 2002–2025. Panel B: desert-elimination (open one office in every desert district), 2008–2025 (pre-agent years degenerate). Shaded regions mark the gap vs observed. Full series in Table A6.

6 Discussion and Conclusion

Between 2002 and 2025, Peru's office footprint expanded from 365 to 533 districts served (19.5 to 28.5 percent), while any-access coverage grew from 613 districts in 2011 (32.7 percent, the first year in which agents had a material presence) to 1,788 districts in 2025 (95.5 percent). Applying the HOI framework, previously used for schooling, water, sanitation, and electricity, reveals a stark two-tier pattern. The office HOI moved only from 0.131 in 2002 to 0.202 in 2025, with all the gain coming from coverage; the D-index behind it stayed flat at ≈ 0.30, so offices did not equalize access over two decades. The any-access HOI, by contrast, jumped from 0.131 in 2002 to 0.937 in 2025, driven jointly by near-universal coverage and a D-index collapse from 0.326 in 2011 to 0.019 in 2025. The Shapley decomposition identifies density and poverty as the market-size variables that sort office location, while the 85 remaining desert districts (4.5 percent of the territory, 0.37 percent of the population) are defined by the Andean and Amazonian geography that even agents have not yet reached.

The remaining inequality lies not in whether districts have access, but in what kind. Two-thirds of Peruvian districts (67.0 percent) are served only by banking agents. The 2025 HOI gap between the two outcome definitions captures this quality gradient in a single number: office HOI at 0.202 versus any-access HOI at 0.937, a difference of 0.735 HOI points. Because offices are located by profit-seeking institutions whose expansion stops where expected revenue no longer covers fixed costs, the office HOI describes an equilibrium of the private banking frontier, not a shortfall against a universality mandate. The geography-only robustness check reinforces this interpretation: altitude, remoteness, and natural region (variables entirely exogenous to market dynamics) account for 77.8 percent of the full HOI penalty.

Five caveats qualify the analysis. First, the logit does not incorporate spatial dependence; access probabilities likely depend on whether neighboring districts have offices, and incorporating a spatial-lag variable is a natural extension. Second, district of residence is not purely a circumstance if individuals migrate; for the small, remote districts that drive the D-index, however, the dominant pattern is outmigration. Third, not all six circumstances are equally pre-determined in the strict Roemerian sense. Poverty, density, and indigenous-language share include components that reflect aggregate historical outcomes of residents' decisions and, in the case of poverty, potentially of the presence of financial infrastructure itself. This is a general limitation of applied IOp analysis, discussed by Roemer and Trannoy [2016]: almost no observable variable is a pure circumstance in the sense of being fully exogenous to aggregate effort and history. The geography-only specification (altitude, remoteness, natural region), which recovers 77.8 percent of the HOI penalty, is the reading of the exercise that survives the strictest interpretation of the circumstance concept. Fourth, without an observed effort variable and without a demonstrated monotonicity property for the binary D-index, the estimates should be read as partial, ex-ante measures of circumstance-related inequality: informative about the six observed circumstances but not a formal lower bound. Fifth, Peru's digital-finance ecosystem, which includes Yape [Banco de Crédito del Perú, 2025], the *Cuenta-DNI* [Congreso de la República del Perú, 2021], and BCRP interoperability [Banco Central de Reserva del Perú, 2022], narrows the service gap for simple transactions but does not substitute for physical access in credit, cash deposits, or complex services, and digital adoption itself is unequally distributed across the same circumstances measured here.

The policy reading is that Peru achieved near-universal basic access through a combination of Banco de la Nación, CMACs, and private agents, but left a quality gradient between territories with full-service offices and those with only agents. Addressing this gap requires either public incentives for office expansion beyond the market frontier, or regulatory upgrades that allow agents and digital platforms to deliver credit, insurance, and advisory services to the two-thirds of districts that lack a physical office. Extensions include applying the framework to other agent-banking models (Colombia, Mexico, Brazil), incorporating district-level data on digital transactions, linking territorial IOp in access to individual IOp in financial outcomes, and using the office plateau as a natural experiment to identify the causal consequences of the quality gap.

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Appendix

Table A1: Coverage, D-index, and HOI: financial offices vs. any access, annual series 2002–2025

	Financial offices			Any access (office or agent)		
Year	Cov.	D	HOI	Cov.	D	HOI
2002	0.195	0.326	0.131	0.195	0.326	0.131
2003	0.201	0.313	0.138	0.201	0.313	0.138
2004	0.202	0.318	0.138	0.202	0.318	0.138
2005	0.205	0.323	0.139	0.205	0.323	0.139
2006	0.210	0.321	0.142	0.210	0.321	0.142
2007	0.213	0.327	0.143	0.213	0.327	0.143
2008	0.221	0.324	0.149	0.255	0.361	0.163
2009	0.232	0.315	0.159	0.284	0.356	0.183
2010	0.241	0.304	0.168	0.310	0.337	0.206
2011	0.247	0.304	0.172	0.327	0.326	0.220
2012	0.254	0.305	0.177	0.350	0.318	0.238
2013	0.260	0.299	0.182	0.444	0.239	0.338
2014	0.266	0.291	0.189	0.523	0.190	0.424
2015	0.271	0.297	0.190	0.627	0.136	0.542
2016	0.274	0.299	0.192	0.685	0.108	0.611
2017	0.277	0.293	0.196	0.789	0.087	0.720
2018	0.280	0.294	0.198	0.815	0.077	0.752
2019	0.282	0.291	0.200	0.849	0.066	0.793
2020	0.282	0.291	0.200	0.860	0.059	0.809
2021	0.284	0.292	0.201	0.879	0.055	0.830
2022	0.283	0.291	0.201	0.904	0.046	0.863
2023	0.282	0.291	0.200	0.910	0.042	0.872
2024	0.281	0.295	0.198	0.943	0.025	0.920
2025	0.285	0.291	0.202	0.955	0.019	0.936

Notes: Cov. is district-level coverage; D is the D-index; HOI is the Human Opportunity Index. Annual series 2002–2025. Corresponding chart in the main text: Figure 1.

Table A2: District characteristics by financial-office access (2002–2025)

Variable	All districts	Never office	Ever office	Diff (Ever−Never)
N	1,873	1,324	549	
Poverty rate (%)	34.2	37.6	25.8	-11.8
Population (2017)	17,170	4,740	47,146	+42,406
Density (hab/km^2)	439	63	1,347	+1,284
Altitude (m)	2,230	2,493	1,594	-899
Distance to dept. capital (km)	89.6	95.2	76.2	-19.0
Indigenous language (%)	33.1	37.3	23.0	-14.3
Coast region (%)	19.8	12.5	37.3	+24.9
Sierra region (%)	63.7	71.8	44.3	-27.5
Selva region (%)	16.6	15.8	18.4	+2.6

Notes: Circumstance variables from the 2017 population census and the 2018 INEI district poverty map, all time-invariant. "Never office" are districts with zero regulated financial offices in every year 2002–2025; "ever office" had at least one office in at least one year. Distance is Haversine distance to the most populous district of the same department (2017 census population).

Table A3: Coverage, D-index, and HOI under population weighting, annual series 2002–2025

	Financial offices			Any access (office or agent)		
Year	Cov.	D	HOI	Cov.	D	HOI
2002	0.637	0.179	0.523	0.637	0.179	0.523
2003	0.648	0.169	0.538	0.648	0.169	0.538
2004	0.661	0.170	0.548	0.661	0.170	0.548
2005	0.675	0.173	0.558	0.675	0.173	0.558
2006	0.692	0.167	0.577	0.692	0.167	0.577
2007	0.701	0.164	0.586	0.701	0.164	0.586
2008	0.717	0.161	0.602	0.763	0.157	0.644
2009	0.732	0.153	0.619	0.792	0.144	0.678
2010	0.747	0.145	0.639	0.809	0.133	0.702
2011	0.757	0.144	0.648	0.820	0.124	0.718
2012	0.766	0.140	0.659	0.833	0.118	0.736
2013	0.771	0.139	0.664	0.873	0.085	0.798
2014	0.776	0.135	0.671	0.906	0.064	0.848
2015	0.782	0.134	0.677	0.931	0.044	0.890
2016	0.790	0.134	0.684	0.944	0.036	0.910
2017	0.789	0.133	0.685	0.969	0.020	0.949
2018	0.794	0.133	0.688	0.974	0.017	0.957
2019	0.795	0.132	0.691	0.981	0.013	0.969
2020	0.795	0.131	0.691	0.983	0.012	0.971
2021	0.797	0.131	0.693	0.987	0.010	0.977
2022	0.796	0.131	0.692	0.990	0.007	0.983
2023	0.795	0.131	0.691	0.992	0.006	0.986
2024	0.796	0.132	0.691	0.995	0.003	0.992
2025	0.796	0.130	0.693	0.996	0.003	0.994

Notes: Both the logit model and the D-index aggregation use 2017 census population weights.

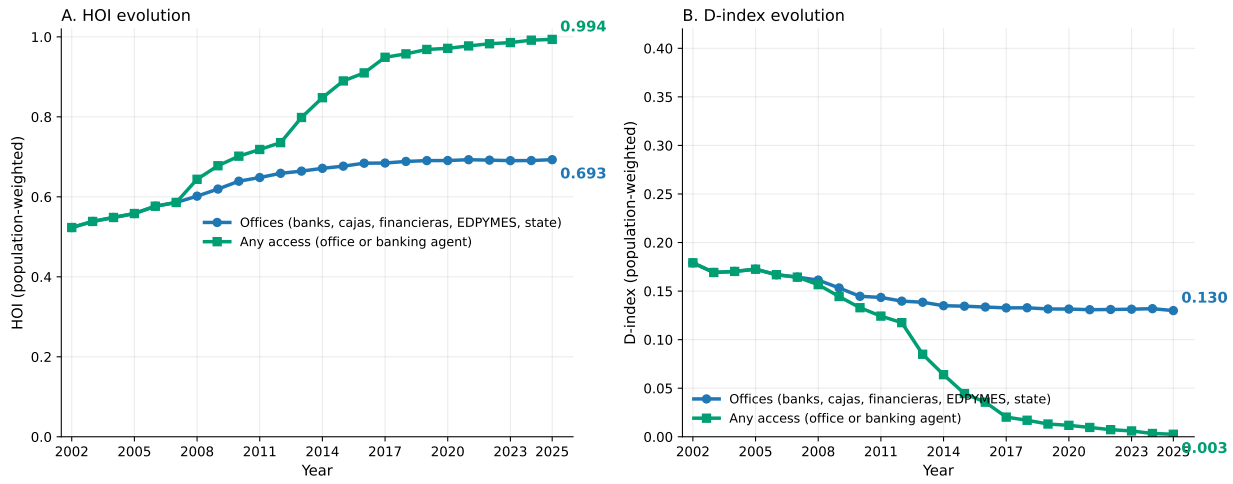


Figure A1: Annual evolution of the population-weighted HOI (Panel A) and D-index (Panel B) under the two access definitions, 2002–2025. Population weights enter both the logit MLE and the D-index formula. In pre-2008 years, has_any equals has_office by construction, so the office and any-access lines overlap.

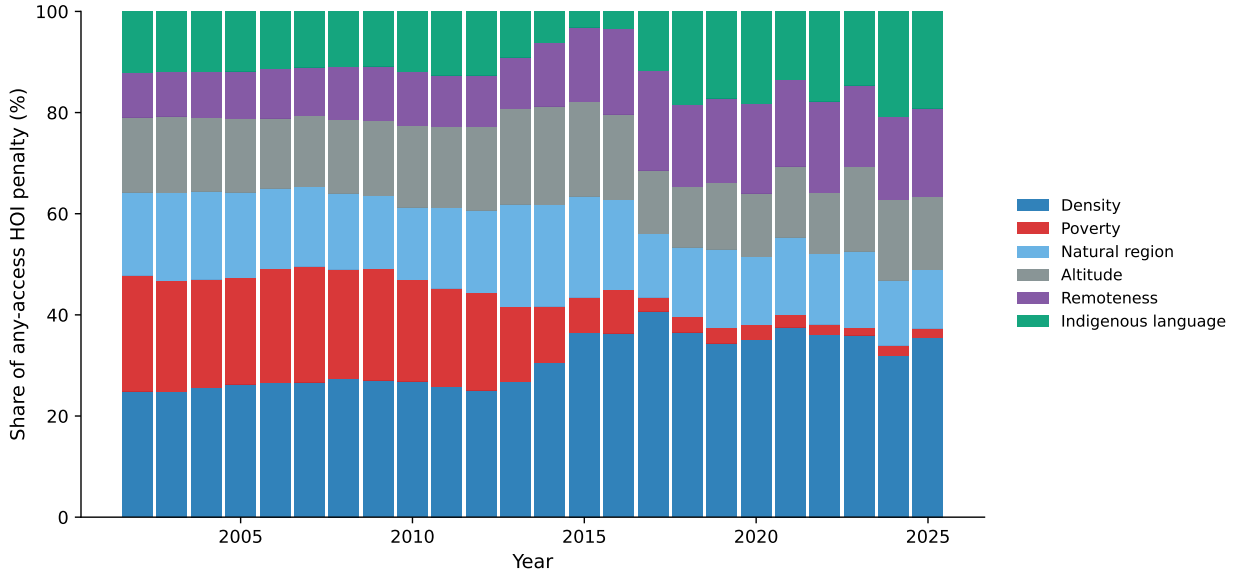


Figure A2: Shapley decomposition of the any-access HOI penalty over time, 2008–2025. To be read with caution: as the any-access D-index collapses toward zero (Figure 1 Panel B), absolute contributions become small and share-based comparisons are noisy. See Section 5 for qualitative interpretation.

Table A4: Year-specific logit models, 2002–2013

Variable	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Constant	-1.700***	-1.693***	-1.718***	-1.712***	-1.671***	-1.659***	-1.609***	-1.540***	-1.467***	-1.418***	-1.413***	-1.360***
	(0.111)	(0.109)	(0.110)	(0.110)	(0.109)	(0.109)	(0.108)	(0.106)	(0.104)	(0.104)	(0.103)	(0.102)
Poverty (z)	-0.436***	-0.390***	-0.386***	-0.387***	-0.431***	-0.460***	-0.439***	-0.444***	-0.426***	-0.414***	-0.405***	-0.425***
	(0.081)	(0.079)	(0.079)	(0.079)	(0.079)	(0.079)	(0.078)	(0.076)	(0.075)	(0.074)	(0.073)	(0.073)
Altitude (z)	-0.189*	-0.154	-0.119	-0.127	-0.112	-0.126	-0.112	-0.136	-0.147	-0.188*	-0.176*	-0.180*
	(0.106)	(0.104)	(0.104)	(0.104)	(0.104)	(0.104)	(0.104)	(0.102)	(0.102)	(0.102)	(0.101)	(0.101)
Density (z)	0.728***	0.699***	0.734***	0.763***	0.770***	0.795***	0.813***	0.787***	0.779***	0.793***	0.816***	0.800***
	(0.084)	(0.082)	(0.083)	(0.084)	(0.083)	(0.084)	(0.084)	(0.083)	(0.081)	(0.081)	(0.082)	(0.081)
Distance (z)	0.207***	0.181**	0.180**	0.165**	0.158**	0.172**	0.157**	0.155**	0.118*	0.112	0.113	0.097
	(0.077)	(0.075)	(0.076)	(0.075)	(0.074)	(0.074)	(0.074)	(0.073)	(0.071)	(0.071)	(0.071)	(0.070)
Coast	0.174	0.336	0.418	0.402	0.350	0.332	0.361	0.376	0.284	0.217	0.309	0.259
	(0.279)	(0.272)	(0.271)	(0.272)	(0.272)	(0.273)	(0.269)	(0.265)	(0.265)	(0.265)	(0.263)	(0.262)
Amazon	-0.126	0.020	0.065	0.133	0.150	0.176	0.176	0.262	0.350*	0.380*	0.488**	0.475**
	(0.230)	(0.225)	(0.224)	(0.223)	(0.221)	(0.221)	(0.218)	(0.213)	(0.209)	(0.207)	(0.205)	(0.204)
Indigenous (z)	-0.209**	-0.180**	-0.187**	-0.186**	-0.167**	-0.153*	-0.173**	-0.136*	-0.109	-0.111	-0.095	-0.081
	(0.088)	(0.085)	(0.086)	(0.086)	(0.084)	(0.085)	(0.083)	(0.081)	(0.078)	(0.077)	(0.077)	(0.076)

Notes: Dependent variable = 1 if district has at least one financial office. Continuous covariates standardized. Sierra is the omitted region. Log-odds coefficients with standard errors in parentheses. Observations = 1,873 for every year. $*p < 0.10$; $**$ $p < 0.05$; $***p < 0.01$.

Table A5: Year-specific logit models, 2014–2025

Variable	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Constant	-1.299***	-1.289***	-1.286***	-1.258***	-1.235***	-1.212***	-1.210***	-1.199***	-1.198***	-1.207***	-1.223***	-1.220***
	(0.101)	(0.101)	(0.102)	(0.101)	(0.101)	(0.101)	(0.101)	(0.101)	(0.101)	(0.101)	(0.101)	(0.100)
Poverty (z)	-0.421***	-0.444***	-0.452***	-0.411***	-0.405***	-0.396***	-0.389***	-0.399***	-0.390***	-0.390***	-0.394***	-0.409***
	(0.072)	(0.073)	(0.073)	(0.072)	(0.072)	(0.071)	(0.071)	(0.071)	(0.071)	(0.071)	(0.071)	(0.071)
Altitude (z)	-0.186*	-0.167	-0.158	-0.190*	-0.203**	-0.232**	-0.237**	-0.236**	-0.247**	-0.245**	-0.222**	-0.208**
	(0.101)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)	(0.102)
Density (z)	0.810***	0.847***	0.867***	0.838***	0.850***	0.835***	0.836***	0.847***	0.842***	0.836***	0.851***	0.830***
	(0.081)	(0.082)	(0.082)	(0.082)	(0.082)	(0.082)	(0.082)	(0.082)	(0.082)	(0.081)	(0.082)	(0.081)
Distance (z)	0.089	0.079	0.079	0.050	0.035	0.033	0.030	0.037	0.041	0.032	0.028	0.028
	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)
Coast	0.195	0.216	0.260	0.266	0.251	0.215	0.209	0.205	0.182	0.183	0.223	0.264
	(0.263)	(0.263)	(0.263)	(0.261)	(0.262)	(0.261)	(0.262)	(0.262)	(0.262)	(0.262)	(0.262)	(0.260)
Amazon	0.471**	0.496**	0.523**	0.496**	0.492**	0.488**	0.484**	0.492**	0.482**	0.496**	0.484**	0.562***
	(0.203)	(0.203)	(0.203)	(0.203)	(0.203)	(0.202)	(0.202)	(0.202)	(0.202)	(0.202)	(0.203)	(0.201)
Indigenous (z)	-0.044	-0.042	-0.034	-0.017	-0.015	-0.012	-0.014	-0.012	-0.015	-0.009	-0.026	-0.007
	(0.074)	(0.074)	(0.074)	(0.073)	(0.073)	(0.072)	(0.072)	(0.072)	(0.072)	(0.072)	(0.073)	(0.072)

Notes: Same specification as Table A4. Log-odds coefficients with standard errors in parentheses. Observations = 1,873 for every year. $*p < 0.10$; $**$ $p < 0.05$; $***p < 0.01$.

Table A6: Policy counterfactuals over time (2008–2025)

	Observed		No-state-entity		Desert elim.	
Year	HOI	D	HOI	D	HOI	D
2008	0.149	0.324	0.087	0.444	0.950	0.017
2009	0.159	0.315	0.091	0.446	0.927	0.023
2010	0.168	0.304	0.103	0.410	0.906	0.027
2011	0.172	0.304	0.105	0.407	0.894	0.028
2012	0.177	0.305	0.114	0.389	0.876	0.031
2013	0.182	0.299	0.120	0.383	0.790	0.032
2014	0.189	0.291	0.123	0.373	0.717	0.035
2015	0.190	0.297	0.118	0.389	0.603	0.064
2016	0.192	0.299	0.121	0.388	0.547	0.071
2017	0.196	0.293	0.125	0.381	0.429	0.120
2018	0.198	0.294	0.125	0.384	0.402	0.135
2019	0.200	0.291	0.126	0.379	0.370	0.145
2020	0.200	0.291	0.126	0.381	0.359	0.150
2021	0.201	0.292	0.126	0.385	0.342	0.157
2022	0.201	0.291	0.126	0.385	0.312	0.175
2023	0.200	0.291	0.127	0.384	0.305	0.179
2024	0.198	0.295	0.125	0.396	0.265	0.213
2025	0.202	0.291	0.132	0.382	0.256	0.226

Notes: The no-state-entity counterfactual removes state-entity offices (BN and Agrobanco) from has_office and re-estimates the logit and D-index. The desert-elimination scenario forces has_office = 1 in all districts with no access point (has_any = 0) and re-estimates. Both scenarios use identical circumstance vectors and the same logit specification, run separately for each year of the panel. Pre-2008 years are omitted because the desert-elimination scenario is degenerate before agents enter the SBS reporting framework: has_any equals has_office by construction, and forcing has_office = 1 in all deserts leaves no variation. Full window (including pre-2008 for the no-state counterfactual) is available in the reproduction files.